

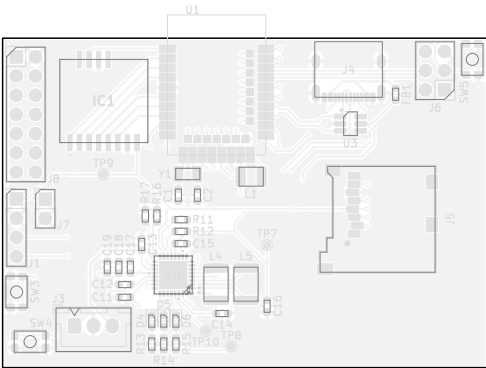
CanSat Devboard Fabrication Document

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
		F,Paste			Paste Mask	
		F,Silkscreen			Legend	GBR
		F,Mask	0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
	Core		1,21mm	FR4	Dielectric	
	Copper	L3 (GND)	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen				Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 °C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY KARPAT SPACE PROGRAM.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

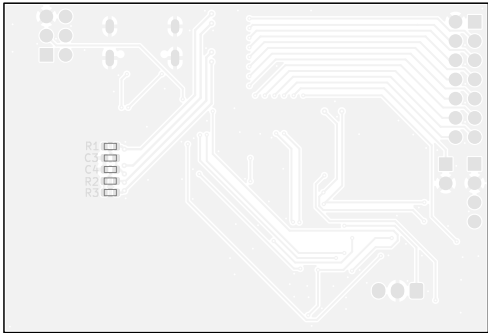
BOARD SIZE	64,059 × 43,561 mm
BOARD THICKNESS	1,660 mm
TRACE WIDTH	0,200 mm
TRACE TO TRACE	0,200 mm
MIN. HOLE (PTH)	0,250 mm
MIN. HOLE (NPTH)	0,650 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,250 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,100 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

		Comments:	Company: Karpat Space Program		Variant: PRELIMINARY	Git Hash: af37c8c
			Board Name: CanSat Devboard		Project Name: Cansat Devboard	
		Sheet Title: Top Fabrication (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
		Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 10

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Bottom Fabrication (Scale 1:1)



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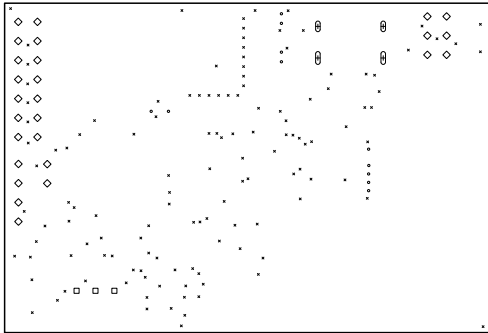
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	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	130	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
○	11	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	3	0,95mm (37,40mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	26	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 174						

Drill Drawing L1 - L4 (Scale 1:1)



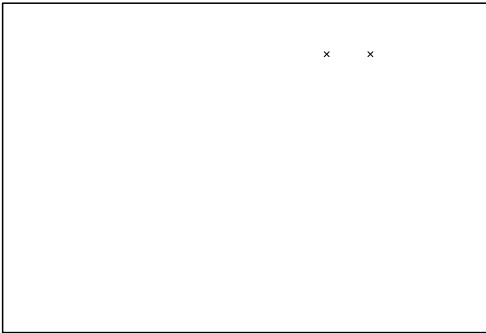
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	Sheet Title: Drill Drawing (L1 - L4)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,65mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
	Total 2					

Drill Drawing L1 - L4 (Scale 1:1)



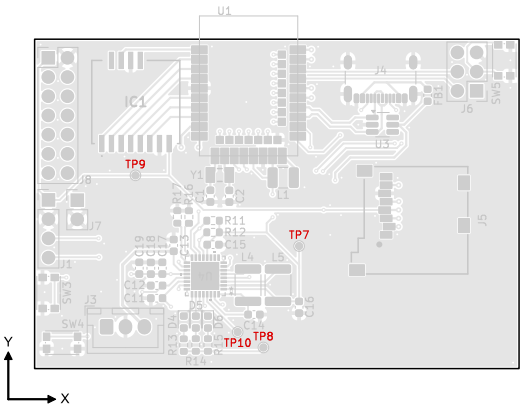
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CanSat Devboard Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP7	VMCU	87.96	1.25
TP8	VSYS	83.23	-12.16
TP9	+3.3V	66.31	10.60
TP10	VLDO	79.83	-10.12

Ref.	Net	X [mm]	Y [mm]
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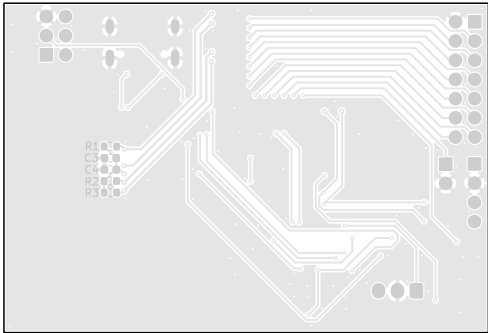
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	Sheet Title: Top Test Points (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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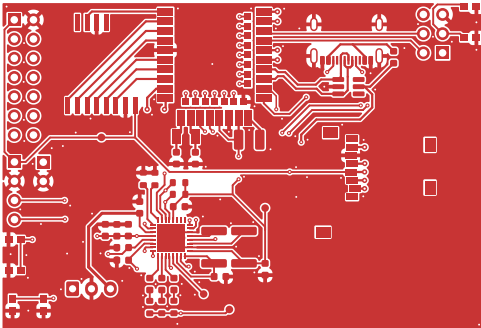


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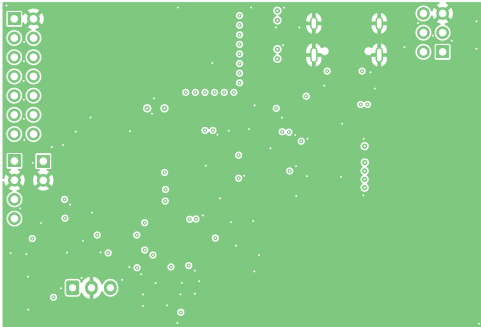
L1 (Sig, PWR) (Scale 1:1)



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		Karpát Space Program		PRELIMINARY	af37c8c
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		CanSat Devboard		Cansat Devboard	
	L1 (Sig, PWR) (Scale 1:1)	File Name:	Designer:	Date:	Revision:
		CanSat_Devboard.kicad_pcb	Bertalan Nagy	2024-04-13	+ (Unreleased)
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L2 (GND) (Scale 1:1)



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		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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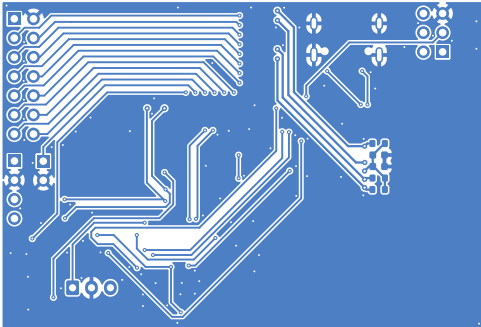
L3 (GND) (Scale 1:1)



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	Sheet Title: L3 (GND) (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



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		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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